

Two Pointers Pattern (DSA Notes)

What is Two Pointers?

Instead of using one index, we use two indexes.

Example:

left = 0

right = arr.length - 1

Array: [10, 20, 30, 40, 50]

L points to the first element and R points to the last element.

Depending on the problem, we move left++, right--, or one of them.

Problem 1: Reverse an Array

Code:

```
int left = 0;
```

```
int right = arr.length - 1;
```

```
while(left < right){  
    int temp = arr[left];  
    arr[left] = arr[right];  
    arr[right] = temp;
```

```
    left++;
```

```
    right--;
```

```
}
```

Problem 2: Valid Palindrome

Compare characters from both ends.

```
while(left < right){  
    if(s.charAt(left) != s.charAt(right)){  
        return false;  
    }  
    left++;  
    right--;  
}  
return true;
```

Problem 3: Two Sum II (Sorted Array)

```
while(left < right){  
    int sum = arr[left] + arr[right];  
  
    if(sum == target){  
        break;  
    }  
    else if(sum < target){  
        left++;  
    }  
}
```

```
    else{  
        right--;  
    }  
}
```

How to Recognize Two Pointers?

Use Two Pointers when you see:

- Sorted array
- Pair of elements
- Reverse array/string
- Palindrome checking
- Remove duplicates
- Compare elements from both ends

Master Template

```
int left = 0;  
int right = arr.length - 1;  
  
while(left < right){  
    // logic  
    left++;  
    right--;  
}
```